

	Dataset	Predict Lengths	ETTh1				Illness			
			Sf-all.	Sf-half.	Ex-half	Masking	Sf-all.	Sf-half.	Ex-half	Masking
Time-LLM [14]	96(24)	Time-LLM	51.8%	5.6%	79.6%	32.5%	99.0%	33.6%	34.9%	64.6%
		w/o LLM	56.0%	4.5%	89.7%	39.5%	76.5%	20.9%	18.4%	53.0%
		LLM2Attn	53.8%	3.3%	92.2%	33.8%	72.7%	20.4%	13.1%	44.6%
		LLM2Trsf	50.3%	3.4%	89.2%	34.8%	74.5%	23.0%	14.3%	49.3%
	192(36)	Time-LLM	43.9%	5.9%	72.1%	32.5%	83.9%	43.4%	30.7%	69.9%
		w/o LLM	46.8%	4.3%	77.3%	32.9%	77.0%	24.5%	15.3%	56.6%
		LLM2Attn	45.0%	4.4%	78.0%	30.0%	73.8%	25.7%	11.2%	53.3%
		LLM2Trsf	44.0%	5.5%	77.1%	29.9%	72.1%	27.0%	6.0%	51.3%
	336(48)	Time-LLM	39.1%	5.9%	67.0%	24.7%	51.6%	26.0%	11.2%	48.5%
		w/o LLM	41.7%	3.9%	71.6%	29.9%	73.6%	27.6%	13.6%	52.9%
		LLM2Attn	34.1%	1.2%	69.5%	21.1%	85.1%	38.8%	12.8%	57.5%
		LLM2Trsf	54.9%	3.7%	75.0%	26.3%	85.2%	42.7%	18.6%	63.6%
	720(60)	Time-LLM	72.6%	31.8%	73.3%	32.1%	73.0%	42.4%	27.5%	76.3%
		w/o LLM	39.7%	5.2%	64.3%	29.3%	74.1%	27.7%	14.2%	56.6%
		LLM2Attn	45.5%	6.0%	66.0%	26.0%	66.2%	28.5%	11.9%	51.7%
		LLM2Trsf	53.3%	10.2%	63.7%	27.3%	67.7%	30.7%	7.8%	49.2%
CALF [21]	96(24)	Dataset	Sf-all.	Sf-half.	Ex-half	Masking	Sf-all.	Sf-half.	Ex-half	Masking
		w/o LLM	50.5%	9.6%	5.6%	8.5%	113.0%	47.4%	24.4%	22.9%
		LLM2Attn	56.2%	12.1%	6.1%	10.4%	118.0%	50.4%	45.8%	28.9%
		LLM2Trsf	51.9%	10.8%	5.8%	7.3%	87.3%	42.4%	35.1%	25.8%
	192(36)	CALF	50.3%	8.5%	5.5%	7.0%	102.6%	56.2%	32.6%	26.0%
		w/o LLM	41.7%	7.8%	3.3%	3.6%	100.9%	45.7%	12.9%	17.1%
		LLM2Attn	50.9%	13.5%	4.5%	6.0%	115.6%	57.0%	28.3%	21.4%
		LLM2Trsf	45.8%	9.7%	3.6%	5.0%	78.8%	41.8%	10.1%	19.9%
	336(48)	CALF	42.4%	8.3%	3.4%	4.3%	73.6%	42.5%	7.0%	17.5%
		w/o LLM	38.1%	9.0%	1.7%	5.0%	68.9%	43.0%	15.1%	14.9%
		LLM2Attn	47.2%	14.5%	2.3%	8.7%	76.3%	49.0%	18.5%	18.1%
		LLM2Trsf	39.3%	10.0%	1.7%	5.5%	78.9%	54.2%	22.2%	15.3%
	720(60)	CALF	40.3%	10.0%	1.9%	4.9%	78.9%	52.0%	23.4%	17.7%
		w/o LLM	36.5%	10.0%	0.7%	5.5%	63.8%	22.7%	5.4%	15.3%
		LLM2Attn	41.0%	10.2%	1.2%	6.2%	69.7%	30.3%	12.1%	19.4%
		LLM2Trsf	36.0%	9.9%	0.7%	5.0%	71.7%	29.2%	12.7%	14.2%
OneFitsAll [49]	96(24)	Dataset	Sf-all.	Sf-half.	Ex-half	Masking	Sf-all.	Sf-half.	Ex-half	Masking
		w/o LLM	62.1%	6.1%	16.6%	31.3%	86.2%	30.9%	36.7%	77.5%
		LLM2Attn	58.6%	6.1%	19.2%	36.1%	68.9%	13.0%	17.3%	43.5%
		LLM2Trsf	68.5%	9.0%	15.0%	34.4%	108.3%	39.8%	44.2%	74.2%
	192(36)	OneFitsAll	58.0%	7.8%	12.6%	30.2%	90.8%	27.4%	40.3%	60.6%
		w/o LLM	52.7%	8.2%	8.8%	28.5%	52.0%	15.2%	8.5%	44.4%
		LLM2Attn	47.5%	6.1%	10.6%	31.8%	76.3%	24.0%	18.0%	47.5%
		LLM2Trsf	80.8%	13.4%	6.7%	25.6%	97.8%	42.0%	36.3%	65.0%
	336(48)	OneFitsAll	54.7%	12.5%	6.4%	24.1%	82.0%	29.6%	24.4%	63.8%
		w/o LLM	62.1%	6.1%	16.6%	31.3%	79.5%	34.8%	25.1%	74.5%
		LLM2Attn	58.6%	6.1%	19.2%	36.1%	78.1%	22.1%	15.7%	49.4%
		LLM2Trsf	68.5%	9.0%	15.0%	34.4%	86.4%	45.8%	21.7%	69.8%
	720(60)	OneFitsAll	58.0%	7.8%	12.6%	30.2%	89.2%	35.8%	21.6%	66.9%
		w/o LLM	40.1%	8.0%	3.5%	25.4%	53.6%	21.8%	9.9%	44.9%
		LLM2Attn	39.4%	7.1%	4.1%	28.2%	74.5%	20.9%	17.1%	45.5%
		LLM2Trsf	93.8%	21.9%	1.5%	25.3%	86.0%	41.4%	27.0%	72.8%

Table 21: Results for input shuffling/masking for Time-LLM, CALF, and OneFitsAll methods on ETTh1 (predict length are "96, 192, 336 and 720") and Illness (predict length are "24, 36, 48 and 60"), the impact of shuffling the input on the degradation of time series forecasting performance does not change significantly before and after model modifications.

Dataset	ETTh2				Electricity			
Input Ablation	Sf-all.	Sf-half.	Ex-half	Masking	Sf-all.	Sf-half.	Ex-half	Masking
Time-LLM	27.1%	4.5%	44.6%	99.5%	212.1%	52.2%	323.4%	103.3%
w/o LLM	31.2%	2.9%	52.4%	137.9%	220.9%	47.3%	332.4%	105.5%
LLM2Attn	30.7%	2.9%	46.7%	134.8%	234.7%	58.2%	350.4%	115.3%
LLM2Trsf	24.8%	2.3%	43.4%	115.0%	240.8%	50.8%	362.0%	118.5%
OneFitsAll	33.0%	1.1%	39.7%	112.1%	237.8%	25.2%	382.0%	88.9%
w/o LLM	32.7%	2.8%	38.7%	115.8%	249.7%	43.4%	384.7%	109.2%
LLM2Attn	27.5%	1.0%	36.5%	137.4%	255.2%	49.5%	393.9%	119.2%
LLM2Trsf	29.6%	2.2%	37.2%	130.8%	263.8%	55.1%	406.4%	125.4%
CALF	11.3%	2.0%	9.1%	44.7%	235.1%	108.2%	29.3%	37.6%
w/o LLM	23.1%	4.1%	17.2%	46.9%	239.5%	109.2%	28.6%	39.4%
LLM2Attn	16.7%	3.4%	14.8%	31.7%	237.9%	116.2%	28.9%	42.2%
LLM2Trsf	18.9%	3.0%	15.3%	43.2%	245.6%	107.0%	29.2%	42.3%

Table 22: For the input shuffling/masking experiments on ETTh2 and Electricity, the impact of shuffling the input on the degradation of time series forecasting performance does not change significantly before and after model modifications.

Dataset	ETTm1				ETTm2			
Input Ablation	Sf-all.	Sf-half.	Ex-half	Masking	Sf-all.	Sf-half.	Ex-half	Masking
Time-LLM	66.6%	10.1%	107.7%	43.5%	47.0%	5.3%	77.7%	202.9%
w/o LLM	68.7%	12.4%	112.0%	52.6%	49.8%	4.7%	78.7%	199.6%
LLM2Attn	73.5%	13.5%	119.3%	50.7%	49.6%	4.7%	75.9%	196.3%
LLM2Trsf	72.0%	27.6%	117.0%	54.2%	46.4%	3.7%	76.5%	191.9%
OneFitsAll	74.4%	8.0%	123.4%	41.8%	48.0%	3.7%	82.9%	172.9%
w/o LLM	66.7%	10.3%	115.1%	45.0%	48.1%	4.4%	81.9%	190.7%
LLM2Attn	73.9%	11.1%	124.8%	51.6%	45.5%	3.3%	77.8%	183.6%
LLM2Trsf	70.1%	8.3%	126.2%	50.5%	48.2%	4.3%	82.0%	182.9%
CALF	64.3%	24.8%	122.3%	13.7%	23.5%	9.1%	47.1%	61.7%
w/o LLM	66.7%	26.0%	123.6%	15.9%	27.1%	11.2%	51.9%	58.1%
LLM2Attn	62.5%	23.4%	122.3%	15.6%	25.2%	8.8%	51.5%	57.3%
LLM2Trsf	61.8%	25.6%	122.6%	13.6%	23.7%	8.2%	51.0%	59.3%

Table 23: For the input shuffling/masking experiments on ETTm1 and ETTm2, the impact of shuffling the input on the degradation of time series forecasting performance does not change significantly before and after model modifications.

Dataset	Weather				Traffic			
Input Ablation	Sf-all.	Sf-half.	Ex-half	Masking	Sf-all.	Sf-half.	Ex-half	Masking
Time-LLM	65.8%	5.5%	85.9%	85.9%	198.1%	56.6%	309.8%	101.3%
w/o LLM	59.2%	4.7%	78.7%	74.6%	196.8%	70.2%	282.1%	78.3%
LLM2Attn	71.8%	8.5%	91.6%	95.3%	212.3%	82.4%	312.7%	92.2%
LLM2Trsf	71.9%	7.1%	94.9%	103.2%	197.5%	64.6%	307.2%	101.1%
OneFitsAll	71.0%	9.8%	102.1%	97.7%	206.4%	11.7%	369.1%	93.5%
w/o LLM	56.4%	2.3%	81.2%	73.5%	238.4%	67.0%	338.6%	85.9%
LLM2Attn	64.9%	5.4%	90.2%	94.8%	222.6%	57.5%	350.5%	95.1%
LLM2Trsf	77.8%	8.3%	101.9%	111.0%	217.8%	35.9%	361.3%	113.3%
CALF	32.6%	4.2%	54.5%	27.2%	201.7%	105.0%	63.1%	27.2%
w/o LLM	28.1%	5.9%	49.4%	13.6%	217.3%	115.4%	67.2%	29.7%
LLM2Attn	26.0%	6.2%	49.2%	17.1%	226.7%	122.7%	69.4%	30.7%
LLM2Trsf	28.0%	6.4%	54.1%	21.1%	232.8%	120.5%	72.0%	29.6%

Table 24: For the input shuffling/masking experiments on Weather and Traffic, the impact of shuffling the input on the degradation of time series forecasting performance does not change significantly before and after model modifications.

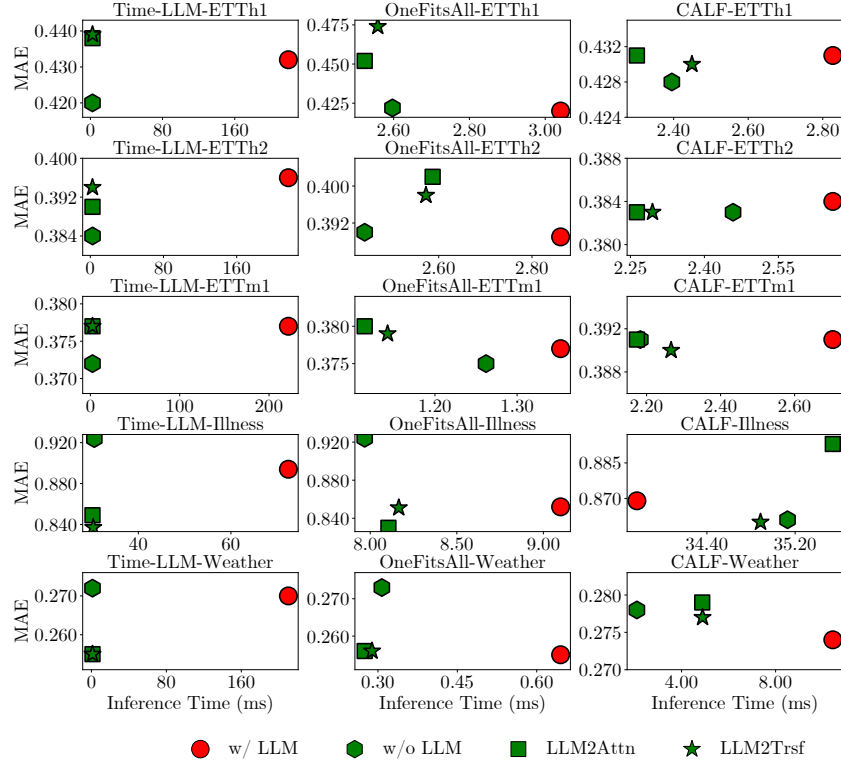


Figure 7: Ablation methods consume less time for inference while providing better forecasting performance in most cases. The figure above shows the inference time and prediction accuracy of Time-LLM, OneFitsAll, and CALF on ETTh1, ETTh2, ETTm1, Illness, and Weather, Traffic datasets, averaged across prediction lengths. Results of other datasets refer to Figure 3.